

We reviewed the Spyder3 Elite back in the May 2009 issue of Digit. That device is for calibrating and colour managing monitors. The two devices for both monitor and printer are packaged together in the Spyder3 Studio, and come with a sturdy metal carrying case – they are both sensitive measuring devices, after all. This package has a recommended price of Rs. 52,000. It will be assumed in what follows that your display has already been calibrated properly. You can also calibrate your camera, but this is not normally done as lighting conditions and colour balance vary so much. Many digital cameras default to a colour space defined as sRGB, and more advanced cameras allow also for the use of Adobe RGB. The information about which is used is embedded within the image file that is created.

So, what does all this mean? If you take a photo of a patch of colour that is known to be primary red, this should be represented

numbers – RGB values – that are associated with them. The profile describes the device's colour gamut, and the associated numerical values.

Using the profile, software can then make adjustments accordingly. To give an over-simplified example, if its profile makes it clear that a particular scanner gives RGB values in which the red is over-emphasised, the software reading the file, something such as Photoshop, perhaps, can adjust the values it receives – it corrects the values that it stores for the peculiarities of the scanner.

When that software then displays the image on screen, assuming that the display device has also been calibrated, it will use the screen profile to adjust the values that it sends to the display. Keeping to a simple example, if the display profile indicates that the display tends to create blues that are too weak, the software will send to the display blue values that are increased by an appropriate amount. A similar process will apply when sending the image to a printer.

By making these adjustments based

on the devices' profiles, the image that is displayed or printed comes much closer to representing the same colours as in the original scanned subject – instead of having reds that are too strong due to the idiosyncrasies of the scanner, and blues too weak thanks to the display.

Of course, it is more complicated than this. The relationship of the numerical values to the range of colours is not simply one such as “over-doing the reds”. The relationship is described either in terms of steps or algebraically. But the biggest problem lies when the gamut of one device is considerably different from another. We'll come back to this point.

The approach taken a few years ago with Windows and general computing devices was really to reduce everything to a lowest common denominator. This meant assuming a rather narrow colour space that just about every device could be expected to cope with, and the use of generic profiles for devices that expressed colours in terms of that colour space. This



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in your computer image as the RGB (red green blue) value of 255,0,0. When that colour value is then sent to a screen for display or to a printer, the colour that results should also be primary red.

There are two main ways in which the digital values and the actual colours imaged or reproduced do not match up: the range of colours – known as the gamut – that can be represented by any device varies, and the accuracy of matching colours to RGB values is off. The way in which this is managed is by the use of profiles. Basically, a profile is simply a file (with extension .ICC or .ICM) that contains a numerical description of the range of colours that a device can handle. It gives the range of colours, together with the

In the process of colour management, the single most tricky aspect is dealing with the printer. This article describes the issues involved, and illustrates the use of printer colour management using the Spyder3Print from Datacolor.

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colour space is sRGB, properly known as sRGB IEC61966-2.1, and was based on the typical characteristics of a cathode ray monitor of a few years ago. This was devised by Microsoft and Hewlett-Packard and is the expected colour space for use on the internet – if you plan to publish any images on the web, always convert them to sRGB before doing so; this should ensure the best results on the widest range of viewing equipment. Typical digital cameras use sRGB by default – but it is worth checking.

It is generally considered that the sRGB colour space includes about a third of all possible real-world visible colours – although the definition of the maximum possible number of colours varies. A third